Given Γ in G with parameterization $\boldsymbol{q}(t)$, $t \in [\alpha, \beta]$, we consider the curve Γ^ε with parameterization $\boldsymbol{q}^\varepsilon$ defined by

$$\boldsymbol{q}^\varepsilon(t) = \boldsymbol{q}(t) + \varepsilon\boldsymbol{\eta}(t), \qquad t \in [\alpha, \beta], \tag{2.3}$$

where $\boldsymbol{\eta}(t)$ satisfies $\boldsymbol{\eta}(\alpha) = \boldsymbol{\eta}(\beta) = 0$. Thus $\boldsymbol{\eta}$ is a C^∞ parameterized curve linking $\mathbf{0}$ to $\mathbf{0}$. In defining variations in this way, we are using, implicitly, the fact that we can add together functions in C^∞ and multiply them by scalars (e.g. ε) and remain in C^∞. We also make use of the intuitive concept that (2.3) defines $\boldsymbol{q}^\varepsilon$ in such a way that it is "close" to $\boldsymbol{q}$ when ε is small. This can be made precise by a little more elaboration, but we forego this here. Effectively, we are using our understanding of C^∞ as a *normed function space* to restrict attention to variations of the base curve Γ in a particular "direction."

Using a Taylor series expansion of the Lagrangian,[3] we have

$$\begin{aligned}\mathcal{A}_L(\Gamma^\varepsilon) - \mathcal{A}_L(\Gamma) &= \int_\alpha^\beta [L(\boldsymbol{q}(t) + \varepsilon\boldsymbol{\eta}(t), \dot{\boldsymbol{q}}(t) + \varepsilon\dot{\boldsymbol{\eta}}(t)) - L(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))]\,\mathrm{d}t,\\ &= \int_\alpha^\beta \left[\varepsilon\left(\frac{\partial L}{\partial \boldsymbol{q}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))\boldsymbol{\eta}(t) + \frac{\partial L}{\partial \dot{\boldsymbol{q}}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))\dot{\boldsymbol{\eta}}(t)\right) + \mathcal{O}(\varepsilon^2)\right]\mathrm{d}t.\end{aligned}$$

Hamilton's principle states that the natural motion of the system described by the Lagrangian L is a stationary point of the classical action which implies that the $\mathcal{O}(\varepsilon)$ term above should vanish for any smooth variation $\boldsymbol{\eta}(t)$ with $\boldsymbol{\eta}(\alpha) = \boldsymbol{\eta}(\beta) = 0$, i.e.,

[3] In the multidimensional setting, Taylor's theorem states that given a C^{k+1} function $f : \mathbb{R}^m \to \mathbb{R}$ and a point z_0, we have

$$\begin{aligned}f(z) - f(z_0) = \nabla f(z_0)\cdot(z - z_0) + f^{(2)}\langle z - z_0, z - z_0\rangle + f^{(3)}\langle z - z_0, z - z_0, z - z_0\rangle\\ + \ldots f^{(k)}\langle z - z_0, z - z_0, \ldots, z - z_0\rangle + \mathcal{O}(\|z - z_0\|^{k+1})\end{aligned}$$

where $\nabla f = \partial f/\partial z$ is the gradient, i.e. a vector with m components, $f^{(2)}$ is the $m \times m$ Hessian matrix of f (the matrix whose ij component is $\partial^2 f/\partial z_i \partial z_j$), and $f^{(2)}\langle \boldsymbol{u}, \boldsymbol{v}\rangle$, $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, represents the quadratic form $\boldsymbol{u}^T f^{(2)}\boldsymbol{v}$. In a similar way we interpret $f^{(3)}$ as a tensor which we can think of as a $m \times m \times m$ triply-indexed array, the ijk element being $\partial^3 f/\partial z_i \partial z_j \partial z_k$ and

$$f^{(3)}\langle \boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}\rangle = \sum_{i=1}^m \sum_{j=1}^m \sum_{k=1}^m (\partial^3 f/\partial z_i \partial z_j \partial z_k) u_i v_j w_k.$$

$$I = \int_{\alpha}^{\beta} \left[\frac{\partial L}{\partial \boldsymbol{q}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))\boldsymbol{\eta}(t) + \frac{\partial L}{\partial \dot{\boldsymbol{q}}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))\dot{\boldsymbol{\eta}}(t) \right] \mathrm{d}t = 0.$$

We use integration by parts to remove the differentiation of $\boldsymbol{\eta}$, thus (in light of the boundary conditions on $\boldsymbol{\eta}(t)$),

$$I = \int_{\alpha}^{\beta} \left[\frac{\partial L}{\partial \boldsymbol{q}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t)) - \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{\boldsymbol{q}}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t)) \right] \boldsymbol{\eta}(t) \mathrm{d}t = 0.$$

Since the variation $\boldsymbol{\eta}$ is meant to be arbitrary, it requires

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{\boldsymbol{q}}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t)) = \frac{\partial L}{\partial \boldsymbol{q}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t)),$$

which is precisely the Lagrangian formulation of the equations of motion.

The method of derivation described here may be formulated in direct analogy to the traditional method of minimizing a function of several variables based on finding critical points. Define the *variational derivative* of a functional $\mathcal{F}$, $\delta\mathcal{F}/\delta\boldsymbol{q}$ so that

$$\mathcal{F}(\boldsymbol{q} + \varepsilon\boldsymbol{\eta}) = \varepsilon \frac{\delta \mathcal{F}}{\delta \boldsymbol{q}} \boldsymbol{\eta} + O(\varepsilon^2),$$

for all suitable (say, C^∞) functions $\boldsymbol{\eta}$. Viewing the action $\mathcal{A}_L(\Gamma)$ as the functional $\mathcal{F}$, the stationarity condition (the Euler-Lagrange equations) may be written

$$\frac{\delta \mathcal{F}}{\delta \boldsymbol{q}} = 0.$$

Thus the calculus of variations becomes a generalization of the traditional method of minimizing smooth functions by finding critical points.[4]

[4]This discussion is a great simplification. Any curve which satisfies this equation will represent a "stationary point" (actually, "stationary curve" would be more accurate) of the classical action. Such curves could include smooth local action minimizers, local action maximizers, or "saddle points" of the actional functional in a generalized sense. Deciding whether a given stationary curve is an actual minimizer of the action would require analysis of the second variation (the coefficient of ε^2 in the expansion above), which introduces additional complexity. For a more comprehensive treatment, see e.g. [210].